# Software Requirements Specification
# for
# Enrich

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# Enrich

## Description

Enrich is a specialized web-based platform designed to streamline the process of English vocabulary acquisition through a flexible and interactive learning environment.

The system provides users with the tools to build a personalized linguistic database by creating individual cards with detailed definitions and organizing them into structured thematic bundles. A key feature of the application is its community-driven ecosystem, where users can share their curated collections and benefit from materials developed by peers. To enhance the training process, the platform integrates a smart generation engine that assembles randomized word sets for practice based on user-defined criteria such as difficulty levels and categories. All activities are monitored by a built-in analytics module that visualizes vocabulary growth and training history through interactive dashboards, ensuring a clear and data-driven learning path.

## Purpose

The purpose of this document is to define the comprehensive software requirements for Enrich. This specification outlines the functional and non-functional requirements to ensure the successful development of a platform dedicated to effective English language acquisition.

The project's primary goal is to create a convenient tool for students and teachers to manage vocabulary through interactive flashcards, community sharing, and automated content generation. This document serves as a technical guide for stakeholders and developers to ensure the final product aligns with specified user needs and system integrity.

## User Classes and Characteristics

| Role | Description |
|---|---|
| Guest | A user who is not logged into an account. They can create a new account (Sign Up) or log into an existing one (Log In). |
| Student | A registered user who creates individual words with definitions, organizes separate cards into themed bundles, and publishes personal collections to the community library. The user browses materials created by peers, imports external bundles into their personal library, and utilizes a smart generator that automatically builds random word sets based on selected categories, tags, or difficulty levels. Additionally, the user views progress statistics, and manages profile configurations. |
| Administrator | A user with full access to content management, including editing words, categories, tags, moderating the global database, and managing user accounts. |

## Subsystems

To organize the functionality, the system is divided into the following subsystems:

- **Account Management Subsystem**: This subsystem handles user identity, access control, and personalization. It processes registration and authentication requests for Guests and manages secure session termination for authorized users. The system also enables the modification of profile data and interface configurations, such as language and visual themes. For Administrators, it provides tools to moderate user accounts by adjusting permissions or restricting access via blocking/unblocking mechanisms.

- **Learning & Content Management Subsystem:** The core subsystem responsible for the lifecycle of linguistic data and its organization. It supports CRUD operations for individual words, allowing users to define translations and linguistic properties. The subsystem facilitates the logical structuring of cards into themed bundles for systematic study. Administrators utilize this module to manage the global dictionary and standardize categories and tags for all users.
- **Community Subsystem:** This module manages the publication of personal word bundles to a shared community library, making them accessible to other students. Users can browse the public repository and import peer-created content directly into their private libraries. The Administrator oversees this subsystem by validating content against safety standards and removing bundles that violate community norms.
- **Smart Training & Analytics Subsystem**: It features a Smart Generator that assembles randomized training sets based on user-defined filters, including categories, tags, and difficulty levels. The analytics component visualizes learning history and vocabulary growth through interactive dashboards and progress diagrams. Administrators have access to audit logs and aggregated platform metrics to monitor system-wide activity.

# List of Use Cases for Subsystems

## Account Management System

| No | Role | Use Case | Description |
|---|---|---|---|
| 1 | Guest | Sign up | A new user creates an account by entering their name, email, and password. |
| 2 | Guest / Admin | Log in | A user authenticates to access the system. |
| 3 | Guest | Verify login data | An automated internal process that checks the database for correct username and password matches. |
| 4 | Guest | Handle authentication error | An exceptional scenario where the system notifies the user of incorrect credentials or non-existent accounts. |
| 5 | Guest | Recover forgotten password | The user initiates a reset process via their registered email if they cannot access their account. |
| 6 | Student / Admin | Terminate active session | The user logs out of the application. |
| 7 | Student / Admin | Modify profile details | The user updates personal information, such as their first name, last name, or contact email. |
| 8 | Student / Admin | Adjust interface settings | The user customizes the application's appearance, including language (Ukrainian/English), visual theme, and font size. |

| | | | |
|---|---|---|---|
| 9 | Admin | Restrict account access | The administrator blocks or deactivates users who violate community safety standards or system policies. |
| 10 | Admin | Restore account access | The administrator unblocks or reactivates previously restricted accounts after review. |

**Learning & Content Management Subsystem**

| No | Role | Use Case | Description |
|---|---|---|---|
| 1 | Student | Create personal words | The user adds new word entries with translations and definitions to their database. |
| 2 | Student | Edit word entries | The user modifies existing translations, definitions, or pronunciation data. |
| 3 | Student | Delete word entries | The user removes specific word cards from their personal vocabulary collection. |
| 4 | Student | Create word bundles | The user organizes selected cards into specific themed collections or bundles. |
| 5 | Student | Edit word bundles | The user modifies the title, description, or card list within an existing bundle. |
| 6 | Student | Delete word bundles | The user removes a themed collection without necessarily deleting the individual words. |
| 7 | Admin | Manage global database | The administrator performs CRUD operations on the shared word library available to all users. |

| | | | |
|---|---|---|---|
| 8 | Admin | Manage categories | The administrator creates, edits, or removes universal categories and tags. |

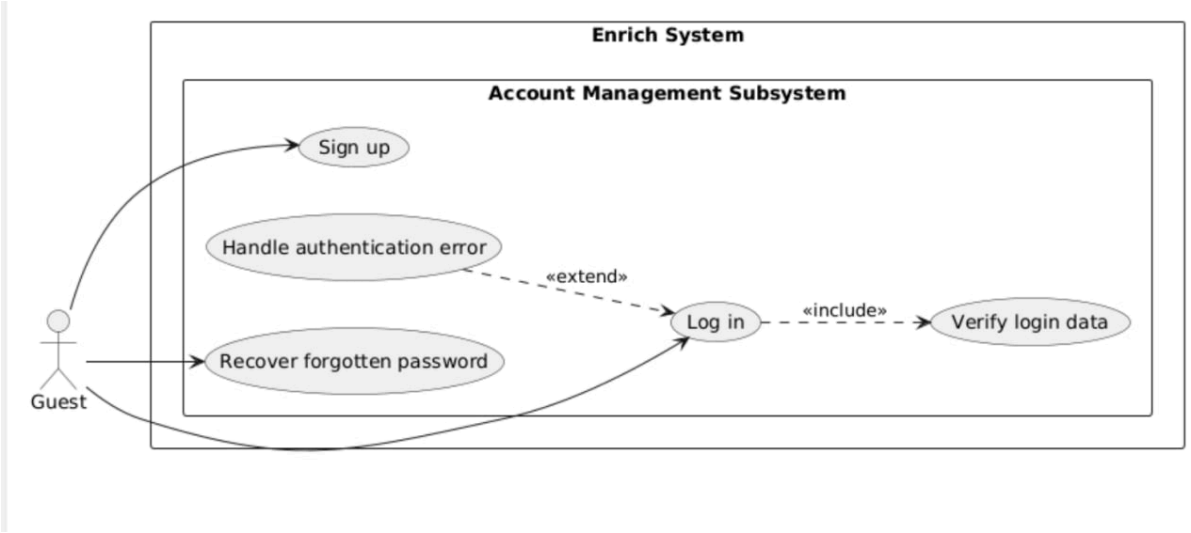
**Community Subsystem**

| No | Role | Use Case | Description |
|---|---|---|---|
| 1 | Student | Publish word bundle | The user exports a personal themed collection to the public community library. |
| 2 | Student | Browse community library | The user searches and views word bundles created and shared by other members. |
| 3 | Student | Import community bundle | The user adds a shared bundle from the community repository into their private library. |
| 4 | Student | Preview community content | The user views the contents and definitions of a bundle before deciding to import it. |
| 5 | Admin | Moderate public bundles | The administrator reviews and removes community content that violates safety or quality standards. |
| 6 | Admin | Feature quality bundles | The administrator highlights high-quality or verified bundles for the community. |

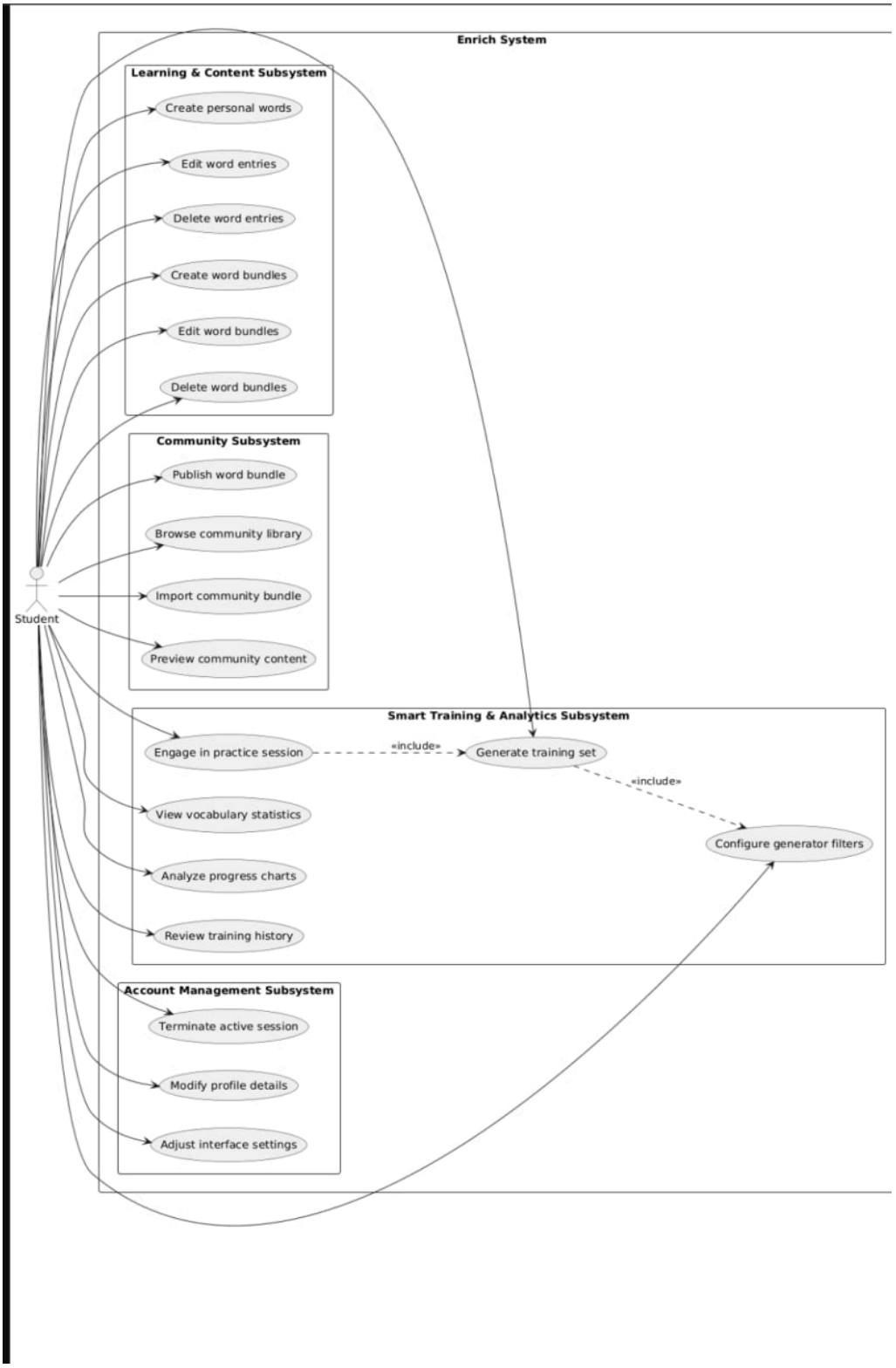
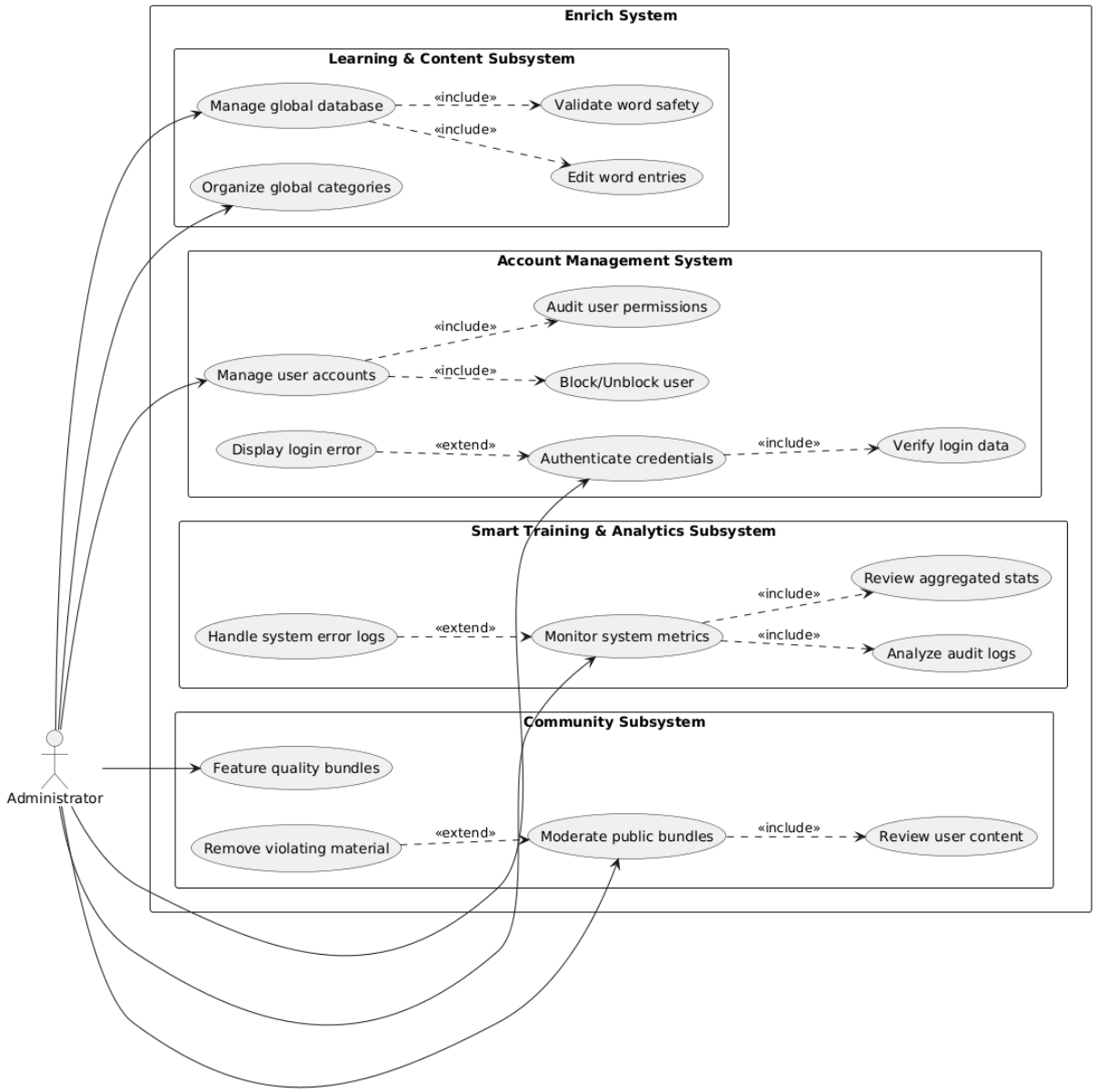
## Smart Training & Analytics Subsystem

| No | Role | Use Case | Description |
|---|---|---|---|
| 1 | Student | Configure generator filters | The user selects specific categories, tags, or difficulty levels to define training parameters. |
| 2 | Student | Generate training set | The system automatically builds a randomized list of words based on the chosen filters. |
| 3 | Student | Engage in practice session | The user interacts with the generated word set to reinforce vocabulary. |
| 4 | Student | View vocabulary statistics | The user reviews the total count of learned words and success rates. |
| 5 | Student | Analyze progress charts | The user examines visual diagrams and graphs showing learning growth over time. |
| 6 | Student | Review training history | The user browses previous results and historical data from past practice sessions. |
| 7 | Admin | Monitor system metrics | The administrator accesses aggregated analytics and platform performance logs. |

# Guest role use case diagram:



# Student role use case diagram:



# Admin role use case diagram: